Similarly, microscopic diatoms living in seafloor sediments secrete chemicals that bind and stabilize the sand. In Canada’s Bay of Fundy, for example, the changes diatoms make to the physical state of the seabed allow other organisms, such as mud shrimp, to colonize it.

British biologists Nancy Harrison and Michael Whitehouse have also suggested that when birds form mixed-species flocks—as many do outside of the breeding season—they are altering their relationship with competitors to find more food resources and gain more protection from predators. The complex social environment they create modifies their own ecology and behavior.

In his explanation of niche construction, Odling-Smee pointed to ancient cyanobacteria, which produced oxygen as a by-product of photosynthesis more than 2 billion years ago. This was a key factor in the Great Oxygenation Event, which changed the composition of Earth’s atmosphere and oceans, massively modifying our planet's environment. The oxygen boost helped create the conditions for the evolution of much more complex life forms—including humans.